Algorithms and Mathematical Concepts for Computing

2nd September 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

Consider the following scenario:

You are tasked with developing a software application for a small business to manage their inventory. The application needs to track product quantities, update inventory levels and generate reports.

a) State in which stage of the software development process are detailed technical specifications and architectural plans created based on the gathered requirements and defined scope? **1**

Mark scheme

Design (1 mark)

b) State which stage of the software development process involves translating the design into a functioning system by writing code and integrating various components. **1**

Mark scheme

Implementation (1 mark)

c) Explain the concepts of abstraction and decomposition in problem-solving and how they can be applied when designing software solutions. You must include an explanation of the purpose and benefits of both abstraction and decomposition. Provide one real-life example for each concept (abstraction and decomposition). **8**

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Mark scheme
Abstraction and decomposition involve breaking down a complex problem into smaller, more manageable parts, focusing on essential details while hiding unnecessary complexities. Abstraction allows us to focus on high-level concepts without getting bogged down in implementation details, while decomposition breaks a problem into smaller, more manageable sub problems, making it easier to understand and solve. In software design, abstraction and decomposition are essential for managing complexity, improving maintainability, and facilitating collaboration among team members.

Criteria:

- ***Abstraction and decomposition involve breaking down a complex problem into smaller, more manageable parts (1 mark)***
- ***Abstraction allows us to focus on high-level concepts without getting bogged down in implementation details (1 mark)***
- ***Decomposition breaks a problem into smaller, more manageable sub problems (1 mark)***
- ***Abstraction and decomposition are essential for managing complexity, improving maintainability, and facilitating collaboration (1 mark)***

- ***Real-life Examples: 2 marks for each example. Below are some examples, but any other real-life example will also be acceptable.***

  1. Abstraction Example: (2 marks)

  Abstraction in a Web Application: A web developer uses a high-level framework like Django or Ruby on Rails to build a web application. The framework abstracts the underlying HTTP requests, database interactions, and session management, allowing the developer to focus on implementing business logic and user interface components.
  Purpose/Benefit: This speeds up development, as developers do not need to deal with low-level details and can instead use predefined components and conventions provided by the framework.

  2. Decomposition Example: (2 marks)

  Decomposition in a Banking System: A banking system might be decomposed into several key services such as account management, transaction processing, loan management, and customer support. Each service handles specific functionalities and can be developed, tested, and maintained independently.
  Purpose/Benefit: This decomposition improves maintainability and allows the bank to update or scale individual services without disrupting the entire system, ensuring better performance and reliability.

Total 10 Marks

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Question 2

Consider the following content related to numbers and number systems:

a) Which of the following is **NOT** a type of number? **1**

i) Integer
ii) String
iii) Natural number
iv) Irrational number

Mark scheme

ii) String *(1 mark)*

b) Decimal to _____ conversion involves converting a number from base 10 to base 8. **1**

Mark scheme

octal (1 mark)

c) Describe the concept of prime factorisation and explain its significance in identifying prime numbers. **2**
Mark scheme

- ***Prime factorisation involves breaking down a composite number into its prime factors, which are prime numbers that multiply together to give the original number. (1 mark)***
- ***This process helps determine if a number is prime or composite based on the number of distinct prime factors it has. (1 mark)***

d) Perform the following conversions: **2**

i) Convert the binary number 1011 to octal number.

Mark scheme:

1011 → 001 011 → 13

Answer: 13 (1 mark for correct answer, 1 mark for working

ii) Convert the octal number 45 to binary number. **2**

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Mark scheme:

45 → 100 101

Answer: 100 101 (1 mark for correct answer, 1 mark for working)

iii) Convert the decimal number 13 to Hexadecimal number. **2**

Mark scheme:

Step 1: Divide the decimal number by 16 and note the remainder:

- 13 ÷ 16 = 0 remainder 13

Step 2: Divide the quotient by 16 and note the remainder:

- 0 ÷ 16 = 0 remainder 0

Step 3: Convert the remainders to hexadecimal:

- 13 = D

- 0 = 0

Step 4: Concatenate the hexadecimal digits: 0D

Answer= 0D or D (both acceptable) (1 mark for correct answer, 1 mark for working)

Total 10 Marks

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Question 3

Answer the following questions related to logic and Boolean algebra:

a) Which of the following operators represents the logical AND operation? **1**

i) ^
ii) &
iii) |
iv) !

Mark scheme

ii) & (1 mark)

b) Complete the truth table for the Boolean expression: **6**

$$(P \vee Q) \wedge (\neg P)$$

P	**Q**	$\neg P$	$P \vee Q$	$(P \vee Q) \wedge (\neg P)$
T	T			
T	F			
F	T			
F	F			

Mark scheme

Accurate completion of the truth table (each correct column = 2 Marks)

Answer:

P	**Q**	$\neg P$	$P \vee Q$	$(P \vee Q) \wedge (\neg P)$
T	T	***F***	***T***	***F***
T	F	***F***	***T***	***F***
F	T	***T***	***T***	***T***
F	F	***T***	***F***	***F***

c) Explain the concept of conditional statements in Boolean algebra and provide an example. **3**

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Mark scheme

Conditional statements in Boolean algebra express the relationship between two propositions where the truth of one proposition, known as the antecedent, implies the truth of another proposition, known as the consequent. In symbolic notation, a conditional statement is represented by "→" or "⇒".

For example:
- Proposition (P): I am hungry.
- Proposition (Q): I will eat.

The conditional statement "If I am hungry, then I will eat" can be expressed as P→Q.

This means that if I am hungry (P), then I will eat (Q).

Criteria:

- ***the relationship between two propositions (1 Mark)***
- ***the truth of one proposition, known as the antecedent, implies the truth of another proposition, known as the consequent (1 Mark)***
- ***For example, accept any different example (1 Mark)***

Total 10 Marks

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Question 4

Answer the following questions related to Predicate Logic, Set Theory and Venn Diagrams:

a) Which of the following symbols represents the existential quantifier in predicate logic? **1**

- ∀
- ∃
- ∧
- ∨

Mark scheme

- ***Answer: ∃ (1 mark)***

b) Explain the concept of set intersection and provide an example **2**

Mark scheme

Set intersection in set theory refers to the operation that yields a new set containing only the elements that are common to both sets being intersected. (1 mark)

For example, let's consider two sets: (1 mark)
A = {1, 2, 3, 4, 5}
B = {3, 4, 5, 6, 7}

The intersection of sets A and B, denoted as A ∩ B, would include the elements that are present in both sets. So, in this case:
A ∩ B = {3, 4, 5}

c) Calculate the complement of the following set: **1**

A = {1, 2, 3, 4} in the universal set {1, 2, 3, 4, 5, 6}.

Mark scheme

{5,6} (1 mark)

d) What is the Cartesian product of two sets and how is it represented? Calculate the Cartesian product of the following sets: A = {4, 5} B = {x, y, z} **4**

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Mark scheme

The Cartesian product of two sets A and B is the set of all possible ordered pairs where the first element comes from set A and the second element comes from set B.

Criteria:
- ***The Cartesian product of two sets A and B is the set of all possible ordered pairs (1 mark)***
- ***where the first element comes from set A and the second element comes from set B. (1 mark)***
- ***The Cartesian product of sets A and B is: A × B = {(4,x),(4,y),(4,z),(5,x),(5,y),(5,z)} (2 marks)***

e) Calculate the union of the following sets:

2

A = {a, b, c}
B = {c, d, e}

Mark scheme

A ∪ B = {a,b,c,d,e}

- ***No duplicates (1 mark)***
- ***Correct union set (1 mark)***

Total 10 Marks

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Question 5

Answer the following questions related to statistics:

a) Discuss TWO (2) of the measures of central tendency and provide examples of situations where they are appropriate or inappropriate to use. **2**

Mark scheme

Below are some Measures of Central Tendency (each set of appropriate and inappropriate use will be 1 mark) (2 marks)

1. Mean:
- Appropriate Use: The mean is appropriate when data are normally distributed or symmetrically distributed around a central value. For example, it is commonly used to calculate the average score of students in a class.
- Inappropriate Use: It may not be appropriate when there are extreme outliers or when the data is skewed, as outliers can heavily influence the mean and misrepresent the central tendency.

2. Median:
- Appropriate Use: The median is appropriate when the data set is skewed or has outliers, as it represents the middle value when data are sorted. For example, median household income is often reported to better represent the average income in an area, as extreme values do not affect it as much as they affect the mean.
- Inappropriate Use: It may not be appropriate when the data set is small, as it does not take into account the actual values of the data points.

3. Mode:
- Appropriate Use: The mode is appropriate when dealing with categorical data or data that can be categorized into distinct groups. For example, it can be used to find the most common type of car owned by households in a neighbourhood.
- Inappropriate Use: It may not be appropriate when there are multiple modes or when the data set is continuous, as it does not provide a meaningful measure of central tendency in such cases.

b) In statistics, _____ is a measure of how spread out the values in a data set are around the mean. **1**

Mark scheme

variance or standard deviation (1 mark)

c) Calculate the median of the following data set: 25, 20, 30, 10, 15, 35. Show your working. **2**

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Mark scheme

To calculate the median of the data set {25, 20, 30, 10, 15, 35}, follow these steps:

1. Arrange the data in ascending order:
{10, 15, 20, 25, 30, 35}

2. Identify the middle value:
- Since there are 6 numbers in the data set, the median is the average of the 3rd and 4th numbers when the data set is ordered.
- The 3rd number is 20.
- The 4th number is 25.

3. Calculate the median:
- Median = (20 + 25) / 2
- Median = 45 / 2
- Median = 22.5

Therefore, the median of the data set {25, 20, 30, 10, 15, 35} is 22.5.

- ***1 mark for arranging the values in ascending order and for identifying the middle value***
- ***1 mark for final result (22.5)***

d) Explain the concept of quartiles in statistics. **2**

Mark scheme

- ***Quartiles in statistics divide a data set into four equal parts. (1 mark)***
- ***The first quartile (Q1) represents the 25th percentile, the second quartile (Q2) represents the median (50th percentile) and the third quartile (Q3) represents the 75th percentile (1 mark)***

e) Calculate the mean of the following data set: 15, 27, 18, 21,12, 24, Show your working. **3**

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Mark scheme

To calculate the mean (average) of a data set, you sum up all the values and then divide the sum by the total number of values.

Here's how to calculate the mean of the given data set:
(15 + 27 + 18 + 21 + 12 + 24) / 6
=> 117 / 6
=> 19.5

So, the mean of the data set is 19.5.

Criteria:

- ***2 marks for showing the steps.***
- ***1 mark for final result.***

Total 10 Marks

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Question 6

Answer the following questions related to Data Structures and Arrays:

a) Which of the following is **NOT** a characteristic of a data structure? **1**

i) Used to organise and store data in a computer system.
ii) A programming language used for data manipulation.
iii) A tool for analysing data patterns.
iv) A graphical representation of data relationships.

Mark scheme

ii) A programming language used for data manipulation. ***(1 mark)***

b) In data structures, a _____ is a linear collection of elements, where each element points to the next one in the sequence. **1**

Mark scheme

Linked list (1 mark)

c) Define the data type "array" and explain its role in storing data. **3**

Mark scheme (3 marks max)

- ***An array is a data structure that stores a collection of elements of the same data type in contiguous memory locations. (2 marks, 1 mark for each point)***
- ***1 mark for any of the points below***
- ***It allows for efficient access to individual elements by indexing. Arrays play a crucial role in storing data as they provide a structured way to organise and manipulate data elements, facilitating operations such as insertion, deletion and retrieval.***
- ***Additionally, arrays enable the efficient processing of large datasets and support various algorithms and data processing techniques.***

d) Write pseudocode to find the largest element in a one-dimensional array. Assume the existence of a function called ***length(arr)*** which returns the number of elements in the array ***arr***. **5**

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Mark scheme

Pseudocode to find the largest element in a one-dimensional array:

```
function findLargestElement(arr):  (function is optional, no mark for function)
    // Initialise the maximum value to the first element of the array
    maxElement = arr[0]                (1 mark)

    // Loop through the array starting from the second element
    for i from 1 to length(arr) - 1    (2 marks)
                                       1 mark for starting index i from 0,
                                       1 mark for length -1
        // If the current element is greater than the maximum value
        if arr[i] > maxElement         (1 mark)
            // Update the maximum value
            maxElement = arr[i]        (1 mark)

    // Return the maximum value
    return maxElement
```

Accept valid alternatives.

Total 10 Marks

Question 7

Answer the following questions related to Data Structures:

a) Which data structure follows the First-In-First-Out (FIFO) principle? **1**

i) Stack
ii) Queue
iii) Linked List
iv) Array

Mark scheme

ii) Queue (1 mark)

b) In a stack, elements are removed from the _____ and added to the _____. **2**

Mark scheme

- ***top (1 mark)***
- ***top (1 mark)***

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c) Consider a queue initially containing the elements 10, 20, 30. Assume 10 is at the front of the queue and 30 is at the back. Perform the following operations one after the other: **3**

I. Dequeue one element.
II. Enqueue the number 40.
III. Dequeue two elements.

Show the final contents of the queue after these operations

Show your working and draw out the queue after each operation.

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Mark scheme

Step-by-Step Operations:

Initial Queue: {10, 20, 30}

Operation I: Dequeue one element
- Dequeue removes the front element from the queue.

After dequeuing one element (10):

Queue: {20, 30}

Operation II: Enqueue the number 40
- Enqueue adds the element 40 to the back of the queue.

After enqueuing 40:

Queue: {20, 30, 40}

Operation III: Dequeue two elements
- Dequeue removes the front elements from the queue. Since we have two dequeues:

After dequeuing two elements (20 and 30):

Queue: {40}

Resulting Queue:

After performing all operations, the resulting queue is {40}.

Summary of Operations:

1. Initial Queue: {10, 20, 30}
2. After Operation I (Dequeue one element): {20, 30}
3. After Operation II (Enqueue 40): {20, 30, 40}
4. After Operation III (Dequeue two elements): {40}

Visualisation:

- Before Operations:
- Front [10, 20, 30] Back

- After Operation I (Dequeue one element):
- Front [20, 30] Back

- After Operation II (Enqueue 40):
- Front [20, 30, 40] Back

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- After Operation III (Dequeue two elements):
- Front [40] Back

(2 marks for showing the steps. Either visualisation or Summary of Operations is acceptable)

So, the resulting queue after these operations is 40. (1 mark for final resulting queue)

d) Provide **ONE (1)** real-world example where each of a stack and a queue data structure would be used. **4**

Mark scheme

- ***Example: Stack***
  Scenario: Browser History
  Usage:
  - ***When navigating through web pages, the back button functionality in browsers uses a stack to store the history of visited pages.***
  - ***Each new page visited is pushed onto the stack. Pressing the back button pops the top page from the stack, allowing users to navigate back through their browsing history in reverse order of visits.***

  1 mark for a relevant scenario and 1 mark for usage.

- ***Example: Queue***
  Scenario: Print Queue
  Usage:
  - ***In a printing system, jobs submitted for printing are added to a queue. The printer processes each job in the order they were submitted (FIFO).***
  - ***This ensures that print jobs are processed fairly and, in the sequence, they were requested.***

  1 mark for a relevant scenario and 1 mark for usage.

  Accept valid alternatives.

Total 10 Marks

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Question 8

Answer the following questions related to algorithm efficiency and searching techniques:

a) Which of the following best describes **Big O** notation in the context of algorithm analysis? **1**

i) The best-case scenario of an algorithm's time complexity.
ii) The average-case performance of an algorithm.
iii) The upper bound on the growth rate of the algorithm's runtime.
iv) The precise execution time of an algorithm for a given input size.

Mark scheme

iii) The upper bound on the growth rate of the algorithm's runtime. (1 mark)

b) Binary search algorithms have a time complexity of _____. **1**

Mark scheme

O(log n) (1 mark)

c) Write pseudocode for a function for a **linear search** algorithm that searches for a target value in a 1-dimensional array. The function takes the target value and array as parameters. It returns the index of the target value or -1 if the target is not found. **6**

You should assume the existence of a function called ***length(arr)*** which returns the number of elements in the array ***arr***.

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Mark scheme

```
Function linearSearch(array, target):
    for i from 0 to length(array) - 1:
        if array[i] == target:
            return i

    return -1 // Indicates target not found
End Function
```

Criteria:

- ***1 mark for function keyword.***
- ***1 mark for parameters.***
- ***1 mark for correct for loop.***
- ***1 mark for correct if statement.***
- ***1 mark for returning the target value index.***
- ***1 mark for returning -1.***

d) Explain why binary search is more efficient than linear search for sorted arrays. **2**

Mark scheme

- ***Binary search is more efficient for sorted arrays because it halves the search space with each comparison, resulting in a time complexity of O (log n). (1 mark)***
- ***In contrast, linear search traverses each element sequentially, resulting in a time complexity of O(n), making it less efficient for large datasets. (1 mark)***

Total 10 Marks

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Question 9

Answer the following questions related to sorting algorithms and recursion:

a) Which sorting algorithm works by repeatedly selecting the smallest (or largest) element from the unsorted portion of the array and placing it in its correct position in the sorted portion? **1**

i) Bubble Sort
ii) Quick Sort
iii) Merge Sort
iv) Insertion Sort

Mark scheme

iv) Insertion Sort (1 mark)

b) State what is meant by recursion in programming. **1**

Mark scheme

A function calls itself directly or indirectly to solve a problem. (1 mark)

c) Fill in the blanks in the pseudocode for the ***quickSort*** function implementing the quick sort algorithm. You can assume the existence of the following functions. **6**

- ***length(arr)***: Returns the number of elements in the array ***arr***.
- ***partition (arr, low, high)***: Partitions the array ***arr*** based on a pivot element and returns the index of the pivot.

```
Function quickSort(arr, low, high):
   if _____:
      # Partition the array
      pivotIndex = _____(arr, low, high)

      # Recursively sort the left and right partitions
      quickSort(arr, low, _____)
      quickSort(arr, _____, high)
   End function
```


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Mark scheme

```
Function quickSort(arr, low, high):
    if low < high:
        # Partition the array
        pivotIndex = partition(arr, low, high)

        # Recursively sort the left and right partitions
        quickSort(arr, low, pivotIndex - 1)
        quickSort(arr, pivotIndex + 1, high)
End function
```

1 mark for using pivotIndex and 1 mark for the correct increment/decrement by 1

Criteria:

- ***First Blank (`if` statement): (1 mark)***
  - Answer: `low < high`

- ***Second Blank (Calling `partition` function):***
  - Answer: `partition` (1 mark)

- ***Third Blank (Recursive call to `quickSort` for left partition):***
  - Answer: `pivotIndex - 1` (1 mark for pivotIndex, and 1 mark for -1, total 2 marks)

- ***Fourth Blank (Recursive call to `quickSort` for right partition):***
  - Answer: `pivotIndex + 1` (1 mark for pivotIndex, and 1 mark for +1, total 2 marks)

d) Explain how recursion applies to the process of calculating factorial numbers. **2**

Mark scheme

- ***Recursion in calculating factorial numbers involves a function calling itself with decreasing values until it reaches the base case, typically when the input is 1, (1 mark)***
- ***and then multiplying those values together as it returns back up the call stack. (1 mark)***

Total 10 Marks

Question 10

Answer the following questions related to tree data structures:

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a) Which of the following best describes a binary tree data structure? **1**

i) A hierarchical structure composed of nodes connected by edges.
ii) A structure where elements are arranged in a linear sequence.
iii) A data structure that stores elements in a hash table.
iv) A structure where elements are stored in a stack.

Mark scheme

i) A hierarchical structure composed of nodes connected by edges. (1 mark)

b) Fill in the missing part of the following statement: **1**
In a binary tree, each node has a maximum of _____ children.

Mark scheme

two (1 mark)

c) Construct a binary search tree from the following sequence of numbers: 8, 4, 12, 2, 6, 10, 14. Ensure that 8 is the root of the tree. **4**

Mark scheme

```
        8          (1 mark)

     /     \
   4         12     (1 mark)
  / \       /  \
 2   6    10  14    (2 marks)
```

d) Using the provided binary search tree (BST) below, demonstrate the steps involved in searching for the value 22. Begin the search at the root node and describe each comparison and movement until the value is found or determined to be absent. **4**

```
          18
        /    \
      10      22
     / \     /   \
    5  12   20    25
   / \       \     \
  3   7       21    30
```


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Mark scheme

- ***1. Compare 22 with the root node value 18. (1 mark)***
- ***2. Since 22 is greater than 18, move to the right subtree. (1 mark)***
- ***3. Compare 22 with the node value 22. (1 mark)***
- ***4. The value 22 is found at the node, so the search is successful. (1 mark)***

Total 10 Marks

End of paper

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Learning Outcomes matrix

Task	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	LO1	Yes
2	LO5	Yes
3	LO2, LO3	Yes
4	LO4, LO6	Yes
5	LO9	Yes
6	LO7	Yes
7	LO7	Yes
8	LO8	Yes
9	LO8	Yes
10	LO7, LO8	Yes

Grade descriptors.

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Explain and use Computational Thinking	Demonstrates no understanding of Computational Thinking and its role.	Provides a basic explanation of Computational Thinking.	Discusses specific examples of Computational Thinking in various contexts.	Applies Computational Thinking to solve complex problems effectively.	Provides critical analysis and innovative approaches to Computational Thinking.
Explain and apply Boolean algebra concepts and notation.	Unable to explain Boolean algebra concepts and notation.	Provides a basic explanation of Boolean algebra concepts and notation.	Draws logic diagrams and writes Boolean expressions accurately.	Applies Boolean algebra concepts proficiently in various computing scenarios.	Offers critical analysis and optimizations of Boolean algebra operations.
Explain and use propositional logic concepts and notation	Demonstrates no understanding of propositional logic concepts and notation.	Provides a basic explanation of propositional logic concepts and notation.	Uses truth tables proficiently to evaluate propositions.	Applies propositional logic effectively in analysing complex statements.	Offers critical evaluation of propositional logic limitations and implications.
Explain and use simple predicate logic statements	Unable to explain simple predicate logic statements.	Provides a basic explanation of simple predicate logic statements.	Writes and interprets predicate logic statements involving quantifiers accurately.	Applies predicate logic statements effectively in real-world scenarios.	Provides innovative applications and thorough analysis of predicate logic statements.
Understand and explain number based system	Unable to understand or explain number-based systems.	Provides a basic explanation of number-based systems.	Performs conversions and arithmetic operations accurately in different number bases.	Applies number-based systems proficiently in solving computational problems.	Offers advanced insights and applications of number-based systems.

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Explain and use set theory and notation	Unable to explain set theory concepts and notation.	Provides a basic explanation of set theory and notation.	Uses set notation proficiently to perform operations and interpret Venn diagrams.	Applies set theory concepts effectively in analysing complex relationships.	Offers advanced interpretations and applications of set theory concepts.
Identify and use common data structures	Demonstrates no understanding of common data structures.	Provides a basic explanation of common data structures.	Applies common data structures proficiently in solving computational problems.	Demonstrates advanced understanding and optimization of data structure usage.	Offers innovative applications and optimizations of common data structures.
Explain and use common algorithms	Unable to explain common algorithms for searching & sorting.	Provides a basic explanation of common algorithms for searching & sorting.	Explains efficiency criteria used to evaluate algorithms effectively.	Applies common algorithms proficiently in solving computational problems.	Offers critical analysis and optimizations of algorithm efficiency.
Understand the fundamental concepts on statistics	Unable to understand or explain key statistical measures and concepts.	Provides a basic explanation of key statistical measures and concepts.	Calculates and uses descriptive statistics effectively.	Applies statistical concepts proficiently in analysing real-world data.	Offers advanced insights and applications of statistical concepts

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